Urban Land Surface Temperature

Urban Heat Island (UHI)

- An urban heat island (UHI) is a metropolitan area that is significantly warmer than its surrounding rural areas due to human activities
- The term "heat island" describes built up areas that are hotter than nearby rural areas

Urban Heat Island (UHI)

- Urban materials such as construction material, roofs, asphalt, concrete and roads absorb more heat from the sun
- The subsequent release of the energy causes urban areas to be warmer compared to the surrounding non-urban areas giving rise to the urban heat effect

Urban Heat Island (UHI)

- Depending on the location, the urban heat island effect can be aggravated by the absence or removal of vegetation, which provides shade and evapotranspiration to cool the air on warm days

Urban Heat Island (UHI)

- The dynamics of the heat island effect are a function of the
  - Time
  - Meteorological conditions
  - Local and urban characteristics

  and consequently,

- These functions could be unique for particular urban areas
- In general, parks, lakes and open areas, appear relatively cooler compared to commercial, industrial or dense buildings

Urban Heat Island (UHI)

- Traditionally, urban heat island effects are observed from the measurement of air temperatures from fixed weather observation stations and sensors mounted on moving vehicles

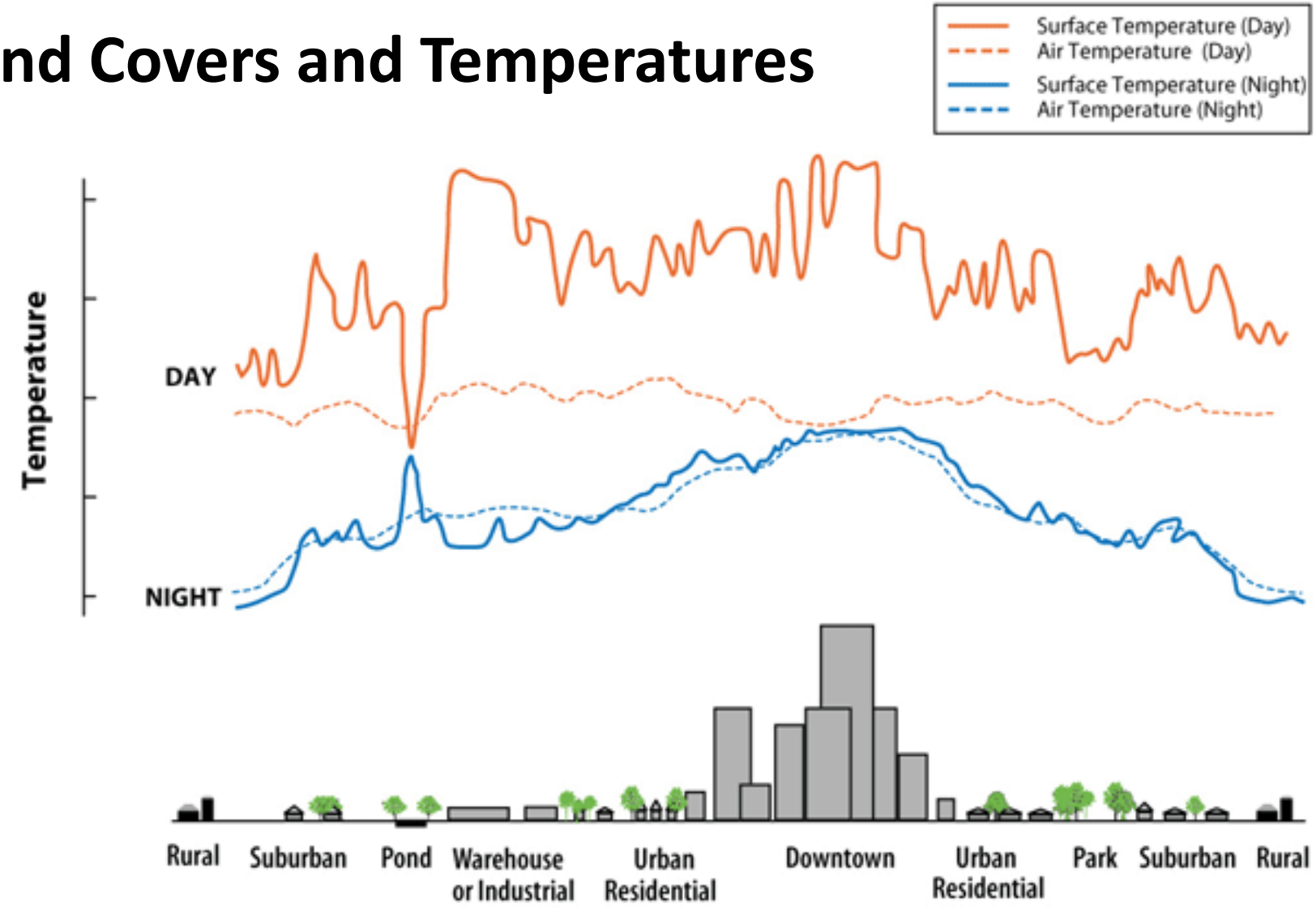
Urban Heat Island (UHI) Effect

- Steps to reduce the heat island effect
  - Increasing tree and vegetative cover
  - Creating green roofs (also called "rooftop gardens“
  - Installing cool—mainly reflective roofs
  - Using cool pavements (permeable pavement parking lot that includes shade trees)

Urban Heat Island (UHI) Effect

- During mornings, urban materials will slowly absorb heat
- Urban heat islands are best observed in the night when maximum radiation from urban materials occurs
- The timing for satellites is therefore not optimized for detecting urban heat islands

Land Covers and Temperatures



Surface and atmospheric temperatures vary over different land use areas. Surface temperatures vary more than air temperatures during the day, but they both are fairly similar at night. The dip and spike in surface temperatures over the pond show how water maintains a fairly constant temperature day and night, due to its high heat capacity.

Urban Heat Island (UHI)

- UHI studies traditionally are conducted with in situ measurements of air temperatures
- Possible to study UHIs both remotely and on continental and global scales
- Remotely sensed thermal imagery has the advantage of providing a time-synchronized dense grid of temperature(Radiant) data over a whole city
- Radiant temperature - what we measure with thermal remote sensing

Land Surface Temperature (LST)

- **Land Surface Temperature (LST) is defined as:**
  - The temperature of the interface between the Earth's surface and its atmosphere
  - The temperature of the air near the Earth´s surface
  - The skin temperature of the ground
- **The LST in remote sensing can be defined as:**
  - The average surface temperature of the ground under the pixel scale(pixel size) mixed with different fractions of surface types

Land Surface Temperature (LST)

- LSTs are regarded as a function of four surface and subsurface properties:
  - Reflectance/ Albedo
  - Emissivity
  - Thermal properties of urban construction materials (including moisture), and
  - The composition and structure of the urban canopy

Land Surface Temperature (LST)

- LST: an important parameter in understanding the urban thermal environment and its dynamics
- Due to the strong heterogeneity of land surface characteristics such as vegetation, topography, and soil, LST changes rapidly in space as well as in time

Land Surface Temperature (LST)

- LST is widely used in a variety of fields including
  - Evapotranspiration
  - Climate change
  - Hydrological cycle
  - Vegetation monitoring
  - Urban climate and environmental studies

Land Surface Temperature (LST)

- Given the complexity of surface temperature over land, ground measurements cannot practically provide values over wide areas
- With the development of remote sensing from space, satellite data offer the only possibility for measuring LST over the entire globe with sufficiently high temporal resolution and with complete spatially averaged rather than point values

Land Surface Temperature (LST)

- Direct estimation of LST from the radiation emitted in the TIR spectral region is difficult to perform with accuracy, since the radiances measured by the radiometers onboard satellites depend not only on surface parameters (temperature and emissivity) but also on atmospheric effects

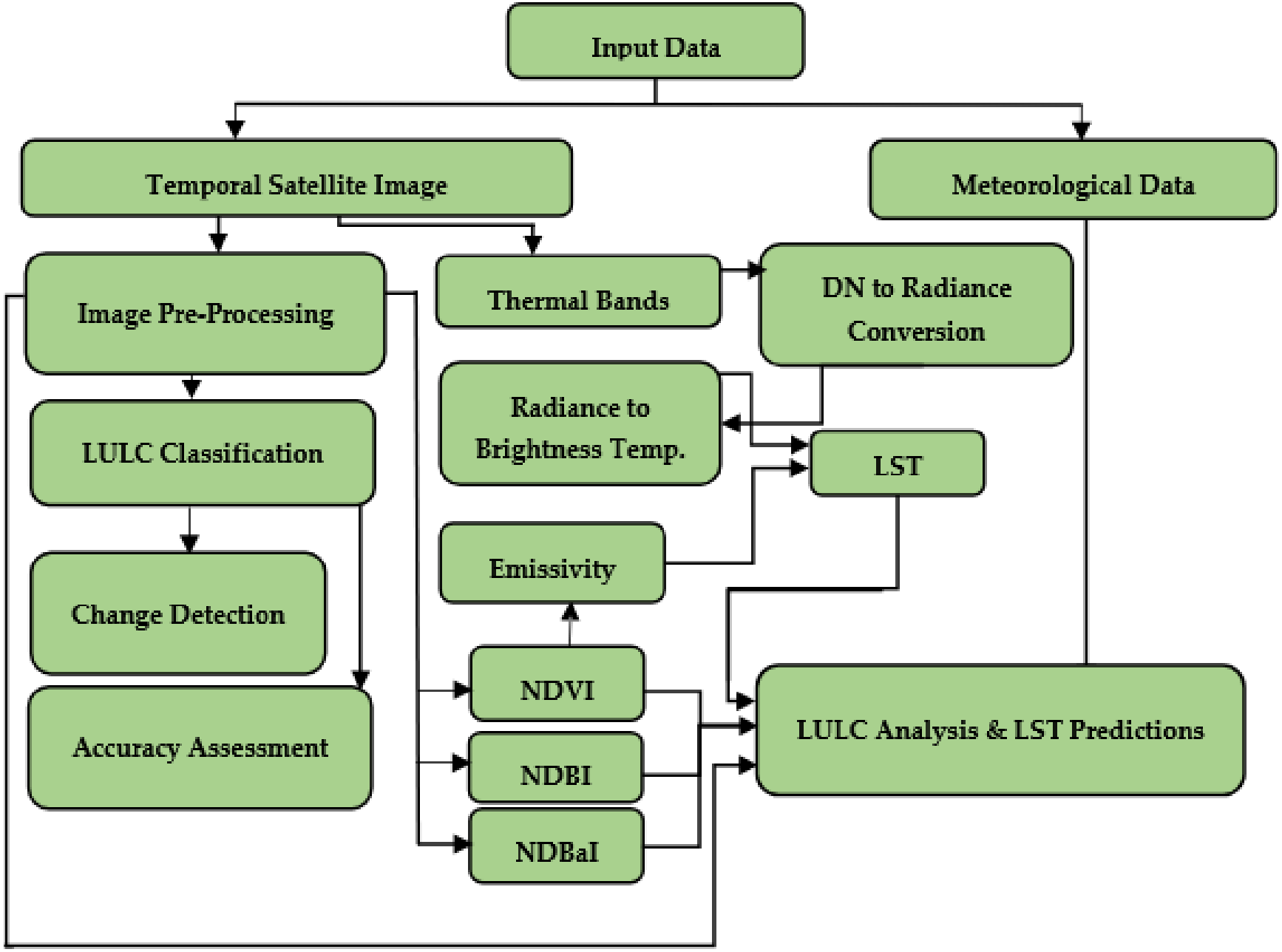
Land Surface Temperature (LST)

- Surface temperature is defined using the radiance emitted by a surface, this temperature also known as the skin temperature

- Brightness temperature is the land surface temperature obtained on the satellite level by thermal infrared (TIR) sensors

Thermal IR Remote Sensing

- Thermal infrared radiation refers to electromagnetic energy within a wavelengths of between 3 and 14 micrometers
- Most remote sensing applications make use of the 3 to 5 and 8 to 14 micrometer range, atmospheric windows (due to minimum absorption)
- Systems on aircraft flying below the ozone layer are not affected and record the full 8-14 µm band


Input Data
Temporal Satellite Image
Meteorological Data
Image Pre-Processing
Thermal Bands
DN to Radiance Conversion
LULC Classification
Radiance to Brightness Temp.
LST
Change Detection
Emissivity
Accuracy Assessment
NDVI
NDBI
NDBaI
LULC Analysis & LST Predictions

Remote Sensing based Land Surface Temperatures

- **Convert DN to Radiance**
- **Calculate at-sensor (top-of-atmosphere) brightness temperature**
- **Calculate NDVI**
- **Calculate Emissivity**
- **Calculate LST**

DN To Radiance

- The DN values range between 0 and 255

$$L_\lambda = \frac{(LMAX_\lambda - LMIN_\lambda)}{QCALMAX - QCALMIN} \times (DN - QCALMIN) + LMIN_\lambda$$

- where the L_{MIN} and L_{MAX} are the spectral radiances for each band at digital numbers 1 and 255.
- DN is the pixel DN value, λ is the wavelength.
- $QCAL_{MAX}$ and $QCAL_{MIN}$ are the pixel values for each band
- L_{MIN} and L_{MAX} values from the header file

At sensor Brightness Temperature

Radiance → Brightness Temperature

Planck's Radiance Function

(At sensor Brightness Temperature) $$T = \frac{K_2}{\ln\left(\frac{K_1}{L_\lambda} + 1\right)}$$

- T is the brightness temperature in Kelvin (K),
- L_λ is the spectral radiance in Wm−2sr−1mm−1,
- and K2 and K1 are calibration constants (Metadata file)

Calculation of Land Surface Emissivity

- **<u>Emissivity can be estimated by utilizing NDVI</u>**
  - **NDVI= NIR-R/NIR+R**

- **Estimation of emissivity**
  - ε_i =1.009 4 +0.04 7 ln(NDVI)

$$LST = \frac{TB}{1 + \left(\lambda \sigma TB / \left(hc\right)\right) \ln \varepsilon}$$

where
λ is effective wavelength,
σ is the Boltzmann constant (1.38 × 10−23 J/K),
h is the Planck's constant (6.626 × 10−34 Js),
c is the velocity of light in a vacuum (2.998 × 10−8 m/sec), and
ε is land surface emissivity.